MILITARY SPECIFICATION

SNOW REMOVAL UNIT, TRUCK MOUNTING MXU-511/E HIGH SPEED CAPACITY

1. SCOPE

1.1 This specification covers one type of high speed, high capacity, truck mounting snow removal unit, designated MXU-511/E.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-C-320	Chromium Plating (Electrodeposited)
QQ-S-781	Steel Strapping, Flat
TT-V-121	Varnish, Spar, Water-Resisting
UU-T-81	Tags, Shipping and Stock
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-C-650	Crates, Wood, Open and Covered
PPP-T-60	Tape, Pressure-Sensitive Adhesive, Waterproof, for Packaging

Military

MIL-C-104	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-P-514	Plates, Identification, Instruction and Marking, Blank
MIL-F-3541	Fittings, Lubrication
MIL-C-3774	Crates, Wood, Open, 12,000- and 16,000-Pound Capacity
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-H-13531	Hose, Rubber and Hose Assembly, Rubber (Hydraulic, Flexible)
MIL-V-13811	Varnish, Waterproofing, Electrical, Ignition
MIL-H-15424	Hand Tools; Packaging of
MIL-T-83118	Trucks, Snow Removal Equipment A/S32U-35 and A/S32U-36 (Heavy Duty)

FSC 3830

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STANDARDS

Federal

FED-STD-151	Metals, Test Methods

Military

MIL-STD-12	Abbreviations for use on Drawings and in Technical-Type Publications
MIL-STD-17	Mechanical Symbols (Other Than Aeronautical, Aerospace, and Spacecraft)
MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-209	Slinging Eyes and Attachments for Lifting and Tying Down Heavy Military Equipment
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods
MS24333	Coupling Assembly, Hydraulic, Self-Sealing, Quick Disconnect, Flared Fitting to Internal Thread Boss
MS24334	Coupling Assembly, Hydraulic, Self-Sealing, Flareless Fitting to Internal Thread Boss

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The snow removal unit shall consist of the following major components:

Item Name	See Requirement
Rotary head assembly	3.6.1
Hydraulic components	3.6.2.

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3.3 Materials. Materials shall be as specified herein. Materials not definitely specified shall be of the best commercial quality and entirely suitable for the purpose. Materials shall be free from all defects and imperfections that might affect the serviceability of the finished unit.

3.4 Design and construction. The snow removal unit shall be designed and constructed as a high speed, high capacity, rotary type assembly for attachment to and functioning with an A/S32U-36 snow removal equipment truck in accordance with MIL-T-83118 equipped with the specified front hitch and, when specified, the A/S32U-10 snow removal equipment truck (see 6.2). The snow removal unit shall connect to the hitch and to the auxiliary engine drive train extending through the hitch. Installation on or removal from the truck shall not require more than 1 man-hour. It shall utilize the full horsepower of the auxiliary engine. All parts necessary for installation on the truck and for efficient and satisfactory operation of the unit when connected to the auxiliary engine power train, hitch, and controls shall be furnished. The unit shall operate without excessive vibration, weaving, rocking, or other action not under complete control of the operator.

3.4.1 The unit shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, and service.

3.4.2 Snow clearance. When mounted on the truck and operating on a smooth level surface at a continuous forward speed of 25 mph, the unit shall clear a swath not less than 10 feet wide at a snow removal rate of 40 tons per minute. The residue left on the surface shall not exceed 1/2 inch in depth.

3.4.3 Center of gravity. The distance from the hitch to the center of gravity of the rotary head assembly shall not exceed 48 inches.

3.4.4 Road clearance. When in the raised position, the snow removal unit shall have a minimum ground clearance of 10 inches.

3.4.5 Reliability. The snow removal unit shall have a minimum MTBF (mean-time-between-failures) of 141 hours with a minimum reliability of 0.84 for a 25-hour mission at a confidence factor of 0.90. For reliability purposes, a failure shall be any malfunction that could prevent satisfactory operation, within the performance limits specified herein, or safe operation in the primary function of snow removal (sheer pin replacement excluded).

3.4.6 Maintainability. The unit shall be designed and constructed as specified herein and to provide the following:

a. Minimum number of parts consistent with reliability and performance specified herein

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b. Minimum amount of training and time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing. Where practical, parts and components shall be located or positioned for rapid and simple inspection and recognition of excessive wear or potential failure

c. Permit adjustments, servicing, replacement of parts and components, and other maintenance with minimum disturbance to other equipment parts or components. Parts and components shall be located for ample and rapid access unless performance or reliability will be appreciably degraded by the accessible location. In performing maintenance, if engineering reasoning or data determines that physical or visual interference between items cannot be avoided, the item predicted to require the most maintenance shall be located for best accessibility

d. Permit maintenance with general-purpose tools and equipment normally available commercially. The use of special-purpose tools and equipment shall be approved by the procuring activity

e. Minimum number of tools required for maintenance by design practices such as reducing the variety of bolt head sizes to the practicable minimum.

3.4.6.1 Foolproofness. Where improper installation of an item could cause malfunction of the item or the equipment in which it is installed, an unsymmetrical mounting means shall be provided where practical. That mounting shall allow the item to be installed only in its proper operating position. If an unsymmetrical mounting means is not practical, the item shall be so mounted that its proper operating position can be readily and visibly determined by service maintenance personnel.

3.4.7 Lifting rings. Lifting rings shall be provided on all major assemblies or components. All lifting rings shall be designed and located in accordance with MIL-STD-209.

3.4.8 Safety guards. Moving parts that constitute a hazard to operating and maintenance personnel shall be fully enclosed or properly guarded. Exposed parts that are subject to high operating temperatures or that are electrically energized shall be insulated, fully enclosed, or adequately guarded. Enclosures or guards shall be provided at the top and sides of the augers and so installed as not to interfere with the snow removal capability of the unit.

3.5 Performance. The snow removal unit, when installed on the A/S32U-36 truck, shall withstand the following conditions without detrimental effect to subsequent operation.

3.5.1 Operation. The snow removal unit shall operate satisfactorily as follows:

a. Remove new fallen snow at a rate of not less than 40 tons per minute at 25 mph

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b. Remove banked or windrowed snow 48 inches deep at a rate of not less than 40 tons per minute

c. Cast a distance of 175 feet to the right and left with cast distance measured to the point of maximum deposit. Assisting wind velocity shall not exceed 5 knots

d. Remove wet snow without clogging in the augers or impellers or bridging in front of the unit.

3.5.2 Endurance. The unit shall perform satisfactorily while operating under sustained usage for at least 40 hours. Servicing during the operating period shall be limited to that specified in 4.6.7. At least 75 percent of the total shall be for periods of not less than 4 hours of continuous operation.

3.5.3 Environmental conditions

3.5.3.1 Operating temperatures. The snow removal unit shall operate satisfactorily in ambient temperatures ranging from -40° to +40°F.

3.5.3.2 Storage temperatures. The snow removal unit shall withstand storage at any temperature ranging from -80° to +160°F.

3.6 Details of components

3.6.1 Rotary head assembly. The rotary head shall have a minimum cutting width of 10 feet and shall discharge snow to the right and left of the vehicle with a minimum of snow blowback. The rotary head shall contain a snow-cutting mechanism suitable for clearing snow 1-1/2 to 60 inches deep. A casting chute shall be provided. The impeller assembly shall be provided with individual or master shear pins or bolts that will protect components of the drive lines and rotary head from damage caused by foreign objects or overload. Shear pins or bolts shall be replaceable by personnel wearing heavy gloves or mittens, without removal of components, cover plates, or shrouding. Shear pins shall be so positioned to permit rapid replacement by personnel standing at the side or located at the rear of the unit. Provisions shall be made to prevent broken shear pins or bolts from being dropped on road surfaces.

3.6.2 Hydraulic components. The hydraulic components shall consist of rams, hoses, piping, fittings, and all other parts essential to practical and efficient operation. The components shall be compatible with the control system on the truck. When additional or special controls are necessary, they shall be provided. A minimum number of fittings, joints, and connections shall be used to prevent undue back pressure, clogging, and leakage. All parts shall be mounted to prevent detrimental vibration and shock. Self-sealing, quick-disconnect couplings conforming to MS24333-8 or MS24334-8 shall be provided to connect to the vehicle junction panel.

3.6.2.1 Flexible hose and hydraulic tubing. Hose for hydraulic lines shall conform to MIL-H-13531, type II, class A. Hydraulic tubing and flexible hose shall be of

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adequate size to permit free flow of hydraulic fluid conforming to MIL-H-5606 at the temperatures specified in 3.5.3.

3.6.2.2 Hydraulic rams. Cylinder walls and rods of hydraulic rams shall be plated with chromium in accordance with QQ-C-320, type II. All seals, rings, gaskets, et cetera, shall be designed for ease of removal and replacement. Piston rams shall be covered or protected to provide for operation throughout the climatic environment.

3.6.3 Casting chute. The casting chute shall be hydraulically controlled through a vertical arc of not less than 150° (75° to either side).

3.6.4 Blade shoes. Alloy-iron, chill cast, adjustable shoes having a through wearing surface hardness of Rockwell C55 to C63 shall be provided and properly located on the unit. The shoes shall be stress relieved to prevent fracturing at temperatures below -30°F. The shoes shall be easily replaceable and shall be adjustable to compensate for at least 1 inch of wear. Shoes shall provide clearance between pavement and rotating snow cutting mechanism. Shoe design shall permit frequent adjustment as necessary to assure cutter ground clearance and that the depth of residue snow does not exceed 1/2 inch. An alternate means of carrying the snow removal unit that will provide for adjustment of cutter ground clearance will be considered only with prior approval of the procuring activity.

3.6.5 Deflectors. Deflectors and necessary devices shall be provided to eliminate hazardous conditions caused by swirling snow that would restrict or impair the operator's view through the vehicle windshield.

3.7 Lubrication. The snow removal unit shall be designed and constructed to operate efficiently and satisfactorily when lubricated with standard military lubricants as specified in MIL-M-8090.

3.7.1 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541 and shall be located in accessible, protected positions. Extended fittings shall be provided to lubricate parts or assemblies that are not readily accessible for direct lubrication or which are likely to be overlooked because of inaccessibility.

3.7.2 Lubrication chart. A lubrication chart shall be provided directing attention to all lubrication fittings and shall specify the range and grade of lubricant required for different temperature ranges. The chart shall be permanently attached in a protected conspicuous location.

3.7.3 Initial lubrication. The unit shall be serviced prior to delivery to the procuring activity.

3.8 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The

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item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.9 Dimensions. The following basic unit dimensions or characteristics shall not be exceeded:

a. Overall width - 126 inches maximum

b. Overall height - (without knives) 60 inches maximum.

3.10 Weight. The weight of the snow removal unit shall be the minimum consistent with the requirements specified herein, and shall not exceed 8,500 pounds.

3.11 Finishes and protective coatings. Cleaning, painting, and finishing shall be in accordance with the best commercial practice. All exterior surfaces shall be yellow, designation DG, conforming to MIL-STD-808. Control handles and other parts subject to wear shall be painted black. Sheet metal surfaces not exposed for finish painting and subject to rusting shall be treated and coated in accordance with the best commercial practices to effectively prevent rusting.

3.12 Markings

3.12.1 Data plates. Transportation data plates conforming to MIL-P-514 shall be provided. All marking plates, placards, charts, instruction plates, et cetera, shall conform to MIL-P-514. The size and shape shall be consistent with the information required thereon. Abbreviations shall be in accordance with MIL-STD-12. Symbols shall be in accordance with MIL-STD-17.

3.13 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.14 Government loaned property. When specified in the contract (see 6.2), the Government will loan one A/S32U-36 or an A/S32U-10 snow removal equipment truck to the contractor upon his request.

3.15 Workmanship. Workmanship shall be of the highest grade throughout and in accordance with good commercial practice for this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

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The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the snow removal unit shall be classified as follows:

a. Preproduction testing . . . . . See 4.4

b. Acceptance tests . . . . . See 4.5.

4.3 Test conditions

4.3.1 Mounting. The snow removal unit shall be mounted on an A/S32U-36 snow removal equipment truck conforming to MIL-T-83118 for all tests. When specified (see 6.2), the A/S32U-10 snow removal equipment truck will be used.

4.3.2 Preliminary run-in. Any preliminary run-in of the unit as may be recommended by the manufacturer shall be performed by the manufacturer prior to submission of the unit for tests.

4.3.3 Data. During the testing specified herein, sufficient data shall be recorded at proper intervals to insure compliance of the snow removal unit with the performance requirements specified. Data shall be recorded every hour during the endurance test. Recorded data shall include the date, time, ambient temperature, average depth of snow, snow density, cast distance, wind velocity and direction, and atmospheric conditions at which the test is conducted as well as data needed to accurately calculate the performance rates, efficiencies, et cetera, to be determined by the specific test.

4.3.4 Test area. The test area shall be approximately 30,000 square feet. Surface under the snow cover shall be smooth level concrete or black top. Snow cover in the test area shall average 4 or over 48 inches deep as required for the particular test. Snow may be windrowed and reused as often as necessary.

4.3.5 Inspection, servicing, and adjustments. Planned stops shall be made at intervals during the endurance test for inspection, servicing, and adjustments. Records shall be kept of all adjustments made regardless of how minor the adjustment. Major adjustments shall be cause for rejection.

4.3.6 Clogging and bridging. During actual plowing operations of 4.6.3, 4.6.6, and 4.6.7, any clogging in the auger(s) or impeller(s) or bridging in front of the unit shall constitute a failure and shall be cause for rejection.

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4.3.7 Production rate. The production rate shall be computed using the following formula:

$$\frac{\text{Production rate}}{\text{(tons per minute}} = \frac{LWDd}{(2000)\ T}$$

Where L = Length of test lane (feet)
W = Width of cut (feet)
D = Average depth of snow (feet)
d = Snow density (pounds per cubic foot)
T = Time to traverse test lane (minutes).

4.4 Preproduction testing (see 6.2)

4.4.1 Test sample. One snow removal unit shall be subjected to the tests specified in 4.4.3.

4.4.2 Test report. After completion of the preproduction tests, a test report shall be prepared in accordance with MIL-STD-831 and three complete copies of the report furnished the procuring activity.

4.4.2.1 Reliability and maintainability information. The following information shall be included as an appendix to the test report:

a. All failures, servicing, adjustments, maintenance, and irregular functioning shall be identified by accumulated operating time, cycles, miles, or position in the test procedure, as appropriate. Test conditions at the time of the events identified shall be recorded

b. A summary of the engineering analysis and of any tests conducted to determine assignable causes for any failure or irregular functioning

c. A summary of the engineering analysis leading to any corrections made to design, construction, quality control, or other procedures, or leading to any corrections to be made or proposed to be made to production items. The summary shall also include an analysis of the predicted effectiveness of such corrections

d. Clock time and man-hours required for each maintenance and servicing action taken during the tests. A brief description of the experience and qualifications of the personnel taking such actions

e. Test activity or contractor comments on item features or requirements that, if modified, should improve the item

f. Test activity or contractor comments on use or maintenance conditions to be avoided or cultivated to increase the reliability or useful life of the item

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g. Any of the above information that is already included in the test report body need not be repeated in the information required by this paragraph, but clear reference to the location of the data shall be included.

4.4.3 Preproduction tests. The preproduction tests shall consist of all tests specified under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual test

b. Sampling plan and test.

4.5.1 Individual test. Each snow removal unit shall be subjected to an examination of product test as described under 4.6.

4.5.2 Sampling plan and test

4.5.2.1 Periodic sampling. One snow removal unit shall be selected at random from each 50 or fraction thereof produced and subjected to the operational test specified in 4.6.3.

4.5.2.1.1 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.5.2.1.2 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a test failure. Final acceptance of the items on hand or produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.6 Test methods

4.6.1 Examination of product. The snow removal unit shall be visually examined for compliance with the requirements of this specification with regard to completeness of assembly, presence of all components and accessories, quality of workmanship and materials, dimensions, hydraulic component leaks, marking, mounting suitability and simplicity, and the functional operation of all components, assemblies, accessories, and controls.

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4.6.2 Mounting and functioning test. The snow removal unit shall be mounted on the truck or a fixture duplicating the truck mounting, controls, pressures, and hydraulic action required to operate all components. All dimensions, fitting locations, and operating clearances shall be checked. All operable components shall be operated to determine that they function satisfactorily. The casting chute shall be reversed from full cast in one direction to full cast in the opposite direction 100 times. The unit shall be raised and lowered 50 times and the unit operated at 50 percent of maximum operating speed for 10 minutes.

4.6.3 Operational test. The truck-mounted snow removal unit shall be operated to determine that it functions satisfactorily as specified herein. Actual plowing operations shall be accomplished in new snow and in banked or windrowed snow for a minimum of 4 hours each to determine that the various components are of adequate strength and function properly. At maximum component operating pressure, no evidence of hydraulic leaking or component failure shall occur.

4.6.4 Hardness. The shoes and wear plates shall be subjected to a hardness test in accordance with FED-STD-151 to determine compliance with the hardness requirements of 3.6.4.

4.6.5 Environmental tests. The snow removal unit mounted on the truck shall be subjected to the following environmental tests in accordance with MIL-STD-810 for storage from -80° to +160°F and operation from -40° to +40°F. After stabilizing at operating temperature, the following operations shall be performed:

a. Rotate the casting chute from the extreme right to the extreme left position and return to extreme right position 100 times

b. Raise and lower the snow removal unit from full down to full up position and vice versa 100 times

c. Operate the unit at 50 percent of maximum operating speed for a minimum of 2 hours

During or after the tests, failure of the rams to fully raise or lower connected components, failure of rotating mechanism to smoothly rotate from one cast position to another, any evidence of hydraulic leakage or component failure, or inability to accomplish the required performance shall be cause for rejection.

4.6.5.1 High temperature. The snow removal unit shall be subjected to high temperature storage and operation in accordance with method 501, procedure I. After stabilizing at operation temperature, the unit shall be subjected to the test specified in 4.6.5.

4.6.5.2 Low temperature. The snow removal unit shall be subjected to low temperature storage and operation in accordance with method 502, procedure I.

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The storage temperature shall be maintained for 48 hours. After stabilizing at operating temperature, the unit shall be subjected to the test specified in 4.6.5.

4.6.6 Snow removal. The snow removal unit shall be subjected to the following capacity and casting tests.

4.6.6.1 High speed. The snow removal unit shall be tested for compliance with the requirement of 3.5.1a. While traveling at a speed of not less than 25 mph, the snow removal rate shall be not less than 40 tons per minute. Mean casting distance, both left and right, measured from the center of the unit to the point of heaviest deposit, shall be not less than 175 feet. Actual performance capability shall be determined by averaging at least four separate operations on a test lane length of not less than 1,000 feet of the test course outlined in 4.3.4. Demonstration of capability to clear snow 1-1/2 inches deep (3.6.1) shall be conducted in conjunction with this test.

4.6.6.2 Deep snow. The snow removal unit shall be tested for compliance with the deep snow requirement of 3.5.1b. Depth of snow shall be not less than 48 inches. Mean casting distance, both left and right, measured from the center of the unit to the point of heaviest deposit, shall be not less than 175 feet. Actual performance capability shall be determined by averaging at least four separate operations on a test lane length of not less than 100 feet of the test course outlined in 4.3.4. Demonstration of capability to clear snow 60 inches deep (3.6.1) shall be conducted in conjunction with this test.

4.6.7 Endurance. The snow removal unit shall be tested to determine its ability to operate under prolonged operating conditions. The unit shall be operated continuously under actual snow removal conditions for periods of not less than 4 hours each for a total of 40 hours, at or near the rated capacity of 40 tons per minute. During this test, the unit's capability of meeting the high speed and deep snow performance requirements shall be determined at the begining and end of each 4-hour period. During these determinations, the rate of snow removal shall be not less than 40 tons per minute. Maintenance and adjustments other than shoe adjustment shall not be permitted during or between the periods of operation. The frequency of adjustment and rate of wear of the shoes shall be entered in the test report and shall include comments with reference to 4.4.2.1 e and f, as appropriate.

4.6.7.1 Servicing. Only the following types of servicing will be permitted during reliability and endurance testing:

a. Truck and auxiliary engine servicing as recommended by the manufacturers

b. Daily lubrication or every 8 hours of operation

c. Such other servicing as recommended and contained in service manuals for this type equipment and for the times specified in such manuals.

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4.6.7.2 Inspection after test. After testing, a thorough inspection of the unit shall be accomplished as follows:

a. Major assemblies, such as the power train components and impeller, shall be disassembled, as necessary, to determine the extent of wear and damage. Failures, erratic operation, excessive noise, heat or vibration suggesting potential failures shall be indicative that disassembly is necessary.

(1) All gears, shafts, bearings, parts, and components disassembled shall be subjected to magnetic particle inspection to determine if cracks have developed through stress.

(2) Castings of disassembled units used to house major assemblies, or used as part of major assemblies, shall be examined under a black light to determine if damage was incurred during the test.

4.6.8 Reliability demonstration and test. Satisfactory completion of all tests specified herein shall be a demonstration of compliance with the quantitative reliability requirements of this specification.

4.6.9 Maintenance, servicing, and unit replacement test. Maintenance tests shall be conducted to determine the adequacy and ease of performing preventative maintenance, servicing, and unit replacement. The frequency of servicing required for the various components shall be determined by study of existing supplier's recommendations and by servicing required during the preproduction tests. Interferences or obstructions to servicing shall be noted. Any tendency of the unit to pick up mud, snow, and ice which interferes with servicing and operation shall be noted. Removal and installation of major components and assemblies to demonstrate ease of replacement shall be accomplished. Particular attention shall be given to time required, difficulties encountered, and the tools and skills required in conducting maintenance, servicing, and replacement.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to verify conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Disassembly. Disassembly shall be the minimum necessary to reduce cubage and safeguard parts known to be subject to damage, pilferage, and loss. Bolts, nuts, washers, and pins shall be placed in one of the mating parts and secured to prevent their loss.

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5.1.1.2 Matchmarking. When necessary to facilitate reassembly, parts that are removed and mating parts shall be matchmarked with lusterless white enamel, overcoated with clear spar varnish conforming to TT-V-121. Small parts shall have matchmarking identification on waterproof, cloth shipping tags conforming to UU-T-81, type A, attached to mating parts.

5.1.1.3 Unpainted surfaces. Unpainted metal surfaces requiring the application of preservation in accordance with MIL-P-116, such as tie rods, adjusting rods, scarifiers, clevises, bolts and nuts, grease fittings, hose coupling and similar parts, shall be coated with type P-1 preservative. Unpainted metal information plates shall be coated with electrical ignition waterproofing varnish conforming to MIL-V-13811.

5.1.1.4 Machined surfaces. Unpainted exterior machined surfaces of component parts shall be coated with type P-2 or P-6 preservative. The coated surfaces shall be wrapped or covered with greaseproof barrier material conforming to MIL-B-121, secured in place with waterproof, pressure-sensitive adhesive tape conforming to PPP-T-60, type III, class 1.

5.1.1.5 Hydraulic systems. The hydraulic fluid supply lines shall be flushed with the approved hydraulic fluid. Wherever practical, pistons shall be fully retracted and any exposed surfaces of the piston rods shall be coated with type P-6 preservative and spirally wrapped with greaseproof barrier material conforming to MIL-B-121, secured in place with waterproof, pressure-sensitive adhesive tape conforming to PPP-T-60, type III, class 1. Open ends of hose shall be sealed with tape conforming to PPP-T-60, type III, class 1.

5.1.1.6 Operating and control levers. Unpainted control levers shall be coated with type P-1 preservative.

5.1.1.7 Extra cutting edges. Unpainted surfaces of extra cutting edges shall be coated with type P-1 preservative.

5.1.1.8 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with MIL-H-15424.

5.1.1.9 Technical publications. Technical publications shall be packaged in accordance with MIL-P-116, method 1C1-1.

5.1.1.10 Intermediate packaging. Component parts removed from each snowplow unit and repair parts shall be intermediate packaged in nailed wood boxes conforming to PPP-B-621, class 2, style optional or in wood cleated-plywood boxes conforming to PPP-B-601, overseas type, style optional.

5.1.2 Level C. The snow removal unit, component parts, tools, and publications shall be preserved and packaged in accordance with the manufacturer's standard practice.

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5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The snow removal unit shall be packed in exterior containers conforming to PPP-C-650, MIL-C-104, or MIL-C-3774. Packaged and unpackaged components shall be placed and secured with flat steel strapping conforming to QQ-S-781 within the shipping container to prevent movement or damage to the components or the snow removal unit.

5.2.2 Level B. The snow removal unit shall be packed as specified for level A, except the ends, sides, and top assemblies shall be omitted and the snow removal unit and component parts shall be shipped on the skid assembly of the container specification selected.

5.2.3 Level C. The snow removal unit, packaged and unpackaged component parts, tools, and publications shall be packed in compliance with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or other carriers rules applicable to the mode of transportation.

5.2.3.1 Blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186.

5.3 Marking. In addition to any special marking required, packages, shipping containers, and the snow removal unit shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

Snow Removal Unit, Truck Mounting MXU-511/E, High Speed Capacity.

6. NOTES

6.1 Intended use. The MXU-511/E snow removal unit is intended for mounting on the A/S32U-10 and A/S32U-36 snow removal equipment trucks for high-speed snow removal from airbase runways and taxiways.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When the snow removal unit is to be designed for use on the A/S32U-10 snow removal equipment truck

c. Whether the A/S32U-36 or the A/S32U-10 truck is required (see 3.15 and 4.3.1)

d. Location and conditions for preproduction testing (see 4.4)

e. Applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Reviewers:
Air Force - 84, 85

Project No. 3830-F115

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

AFLC–WPAFB–OCT 67 2M

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